

Chapter 106

Layer Nodes

This chapter details the use and modification of multilayer images in your compositions.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

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Layer Muxer [LMx]



The Layer Muxer node

Layer Muxer Node Introduction

This tool allows layers from one source to be combined with another source. The tool has two inputs; when multilayer sources are connected, layers from one source are combined with the other.

Inputs

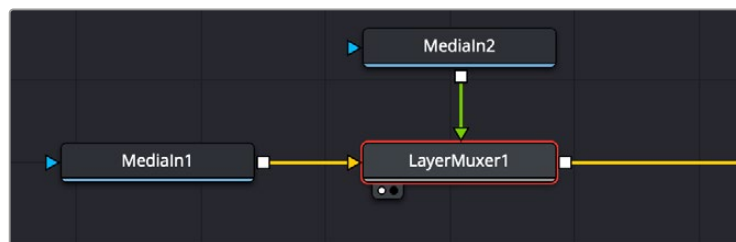
The Layer Muxer tool uses two inputs.

Image 1: This orange input is connected from a node with multilayer data that you want to combine.

Image 2: This green input is connected from a node with multilayer data that you want to combine.

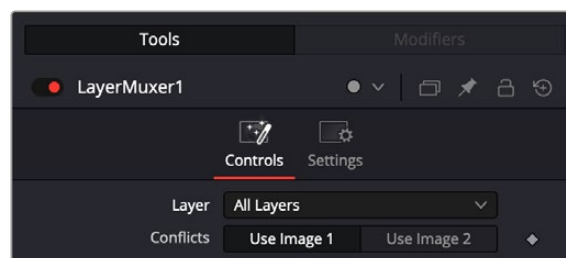
Basic Node Setup

The Layer Muxer node receives two nodes with multilayer data and combines them in various ways based on which layers you want to keep and which you want to ignore.



Layer Muxer combines two multi layer images.

Inspector



Layer Muxer controls

Controls Tab

The Controls tab includes parameters for selecting layers to be used.

- **Layer:** Elect layers from the clip connected to the Image 2 input to combine with the clip connected to Image 1. The pulldown control allows you nominate which layer(s) you'd like to combine. If any option other than All Layers is selected, a small layer icon will appear in the Layer Muxer node to remind you that only certain layers are being passed through.
Default Layer: If a default layer is used in the Input 2 multilayer media file, it will combine that layer.
All Layers: Combines all the layers of the Input 2 multilayer image together.
Custom: Presents a check list off all layers in Input 2 that allows you to turn on and off the multiple layers.
- **Conflicts:** Use this control to switch between either Image 1 or Image 2, the two inputs on this tool; the selected Image will be output as the tool's Default/[Main] layer.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Layer nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Layer Regex [LRx]



The Layer Regex node

Layer Regex Node Introduction

This tool uses complex expressions to filter out layers of multilayer files based on layer naming. Once a filter has been defined, layers that meet the "expression" can be renamed, kept, or removed.

Inputs

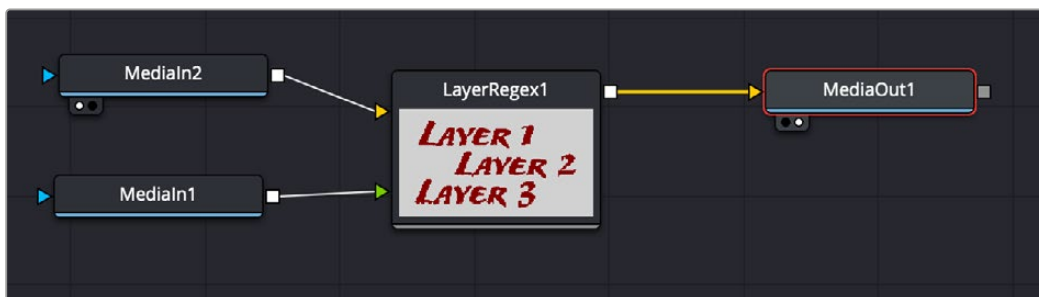
The Layer Regex tool uses two inputs.

Image 1: This orange input is connected from a node with multilayer data that you want to filter.

Image 2: This green input is connected from a node with multilayer data that you want to filter.

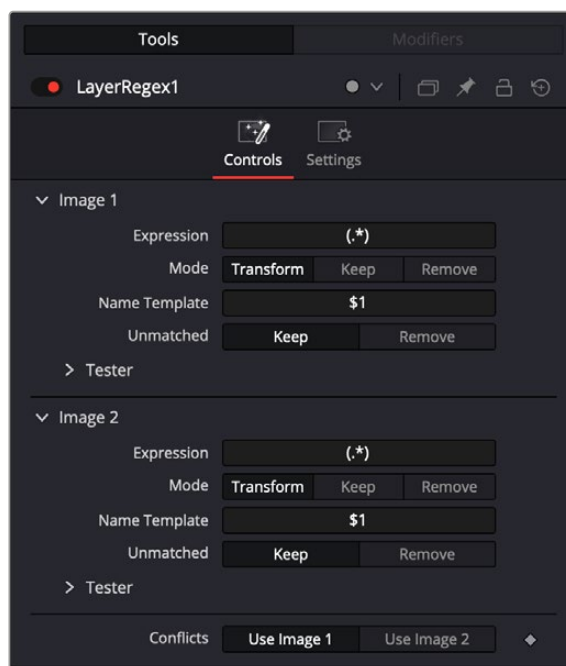
Basic Node Setup

The Layer Regex node receives two nodes with multilayer data and lets you filter them using Regex based on the layer names.



Layer Regex filters layers from two multi layer images.

Inspector



Layer Regex controls

Controls Tab

The Controls tab includes parameters for selecting layers to be used.

- **Expression:** Is the filter used for matching/locating characters.
- **Mode:** Lets you adjust how the expression will be applied.
 - Transform:** Is the act of renaming/replacing characters.
 - Keep:** Keeps the layers specified in the Expression rule.
 - Remove:** Removes the layers specified in the Expression rule.
- **Name Template:** Is the rule applied to the "Expression," allowing you to replace characters or rename layers.
- **Unmatched:** Determines how to deal with layers that don't match the expression.
 - Keep:** Any layers that don't match the Expression are kept; all the layers remain.
 - Remove:** Any layers that don't match the Expression are removed. This isolates the layers that are impacted by the expression.

- **Tester:** A simple tool to view the results of your expression as you change the parameters.
- **Conflicts:** Use this control to switch between either Image 1 or Image 2, the two inputs on this tool; the selected Image will be output as the tool's Default/[Main] layer.

Examples

Add a prefix to existing layer names

Expression: `(.*)`

Name Template: `Right.$1`

This will add `Right.` at the beginning of each of the layer names, giving you `Right.Layer1` , `Right.Layer2`

NOTE: To add a suffix, change the order in the Name Template, `$1.Right.` This will give you `Layer1.Right` , `Layer2.Right`

Find specific characters/words in the existing layers names, then add a prefix

Expression: `(Right.*)`

Name Template `Stereo-$1`

This will locate the layers which contain the word `Right`, then add the word `Stereo-` at the beginning of each of the layer names, giving you `Stereo-Right.Layer1`

Find and replace

Expression: `right(.*)`

Name Template: `Left$1`

This will locate any layer names containing `right` and will then replace that layer name with `Left`. So, layers labeled `right.1` , `right.2` , `right.3` will be renamed to `Left.1` , `Left.2` , `Left.3` .

Keep in mind, when using this method, any layer name that contains the letters `right` (e.g., `copyright`) will be impacted by the expression and will be renamed. To specify a string of characters, use the following expression `^right(.*)`

NOTE: The `^` means match the start of the text. That will make it filter `right` but not `copyright` because it has a `copy` at the start.

Find either word, then remove (or keep)

Expression: `(copy|right)`

Mode: `Remove`

(copyright) locates **copy** and **right** , when the mode is set to Remove, layers containing copy or right, are then removed from the layer list. This expression can be useful for locating variations of a word, for example **(copy|Copy|COPY)**

NOTE: Setting the mode to **Keep** will retain layers that meet the Expression. Using the above example, if the Mode is set to Keep, layers containing copy or right will remain and all other layers are removed.

Find characters, ignoring case

Expression: **(?i)(right)**

This will capture all the characters in the word right, even if they are upper case. This means any capitalized variation of the word right: **Right**, **RIGHT** or **riGht** will be captured.

RegEx Cheat Sheet

For more information on using RegEx <https://www.debuggex.com/cheatsheet/regex/pcr>

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Layer nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Layer Remover [LRm]



The Layer Remover node

Layer Remover Node Introduction

This tool provides a simple way to disable/remove layers from a connected source.

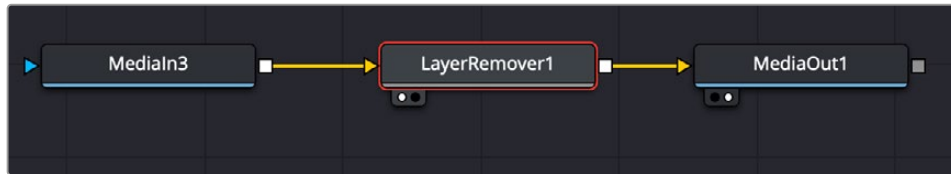
Inputs

The Layer Remover tool uses one input.

Image 1: This orange input is connected from a node with multilayer data that you want to remove a layer from.

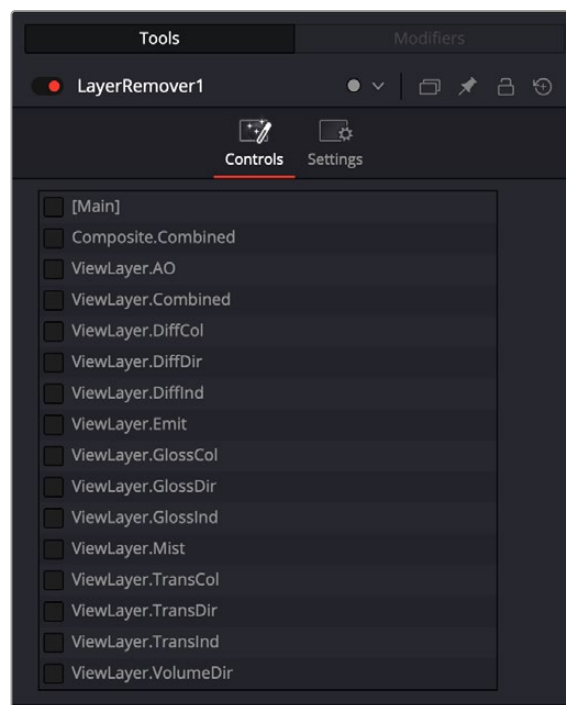
Basic Node Setup

The Layer Remover node receives a node with Multilayer data and lets you strip out specific layers before passing on the image.



Layer Remover lets you disable specific layers of a multilayer file.

Inspector



Layer Remover controls

Controls Tab

The Controls tab presents a simple checklist of all the layers in the multilayer clip. Checking the box will disable/remove layer(s) from the layer list.

Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Layer nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Swizzler [Swz]



The Swizzler node

Swizzler Node Introduction

This tool enables you to combine one or more image layers and/or channels into a custom-made layer or layers. Similar to Channel Booleans, Swizzler can assign channels from a source and apply them to a destination channel, however, the biggest point of difference is that Swizzler utilizes layers, either targeting layers as a source or creating new layers.

To put it simply, this tool has the functionality to create new layers, push source layers or channels to a layer, or remove channels from a source.

Inputs

The Swizzler tool uses multiple inputs.

Input 1: This orange input is connected from a node with image data that you want to combine into a new layer.

Input X: The white inputs are connected from additional nodes with image data that you want to combine into a new layer.

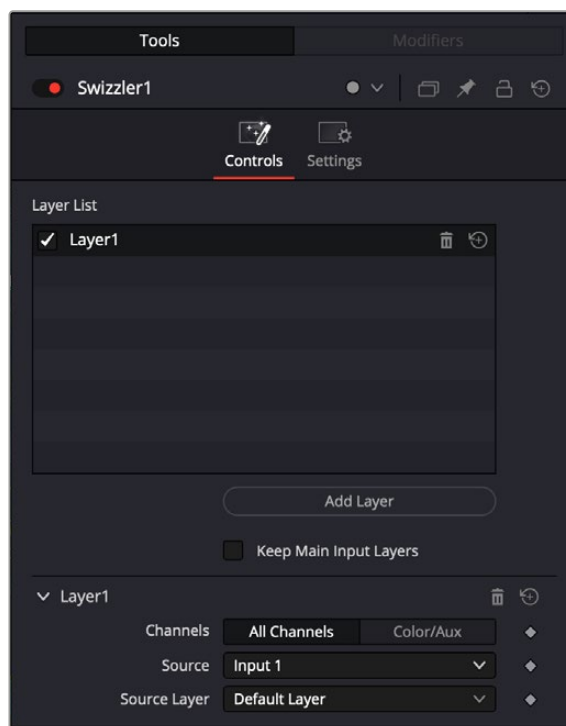
Basic Node Setup

The Swizzler node receives multiple nodes with image data and combines them into new layers.



Swizzler combines multiple images into new layers.

Inspector



Swizzler controls

Controls Tab

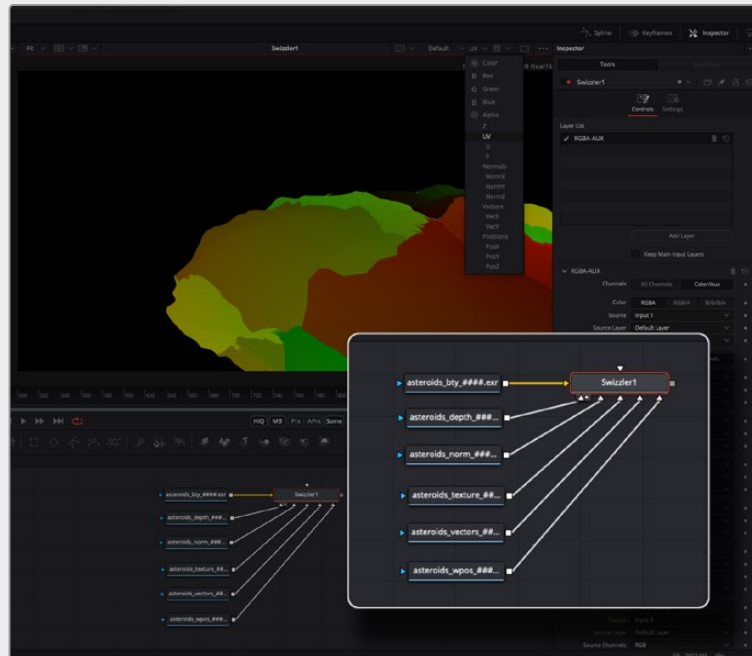
The Controls tab includes parameters for selecting layers to be used.

- **Layer List:** The layer list allows you to view and select custom layers. The button below allows you to add a new layer to this list, introducing a new custom layer. For any layer selected in this list, the options below allow you to choose which channels to include in the layer. Additionally, you can rename layers directly in this list.
- **Add Layer:** This creates a new layer, adding it to the layer list.
- **Keep Main Input Layers:** Checking this box preserves existing layers from the source clip connected to Input1. Having this enabled will pass the layers from Input1 to the Swizzler output, giving you the control to add more layers to an existing multilayer source.
- **Channels:** This control allows you to select which channels will be added to the selected layer in the layer list.
- **All Channels:** Transfers all channels into the selected layer.
- **Color/Aux:** Reveals an additional "Color" and "Aux" options, separating RGBA from AUX channels. This separation allows you to nominate a source for the RGBA and a source for the AUX channels, ultimately combining RGBA from one input and AUX from another input. The other two options, "RGB/A" and "R/G/B/A," provide even further granular controls for channels, expanding the source options for individual channels.
- **Source:** Select an input from this drop-down to source layers or channels for your chosen layer. Your chosen layer will be the one selected in the layer list above.
- **Source Layer:** When using a multi-layer file connected to one of the inputs, you can choose a specific layer using this control. All the available layers from the Source Input control will be displayed.
- **Source Channels (Color/Aux mode):** Select which channels from the chosen layer you would like to use for the Aux channel. If the chosen layer doesn't have Aux channels and has been rendered with only RGB, then select the color channels (i.e., RG for UV Texture or RGB for XYZ Normal).

Examples

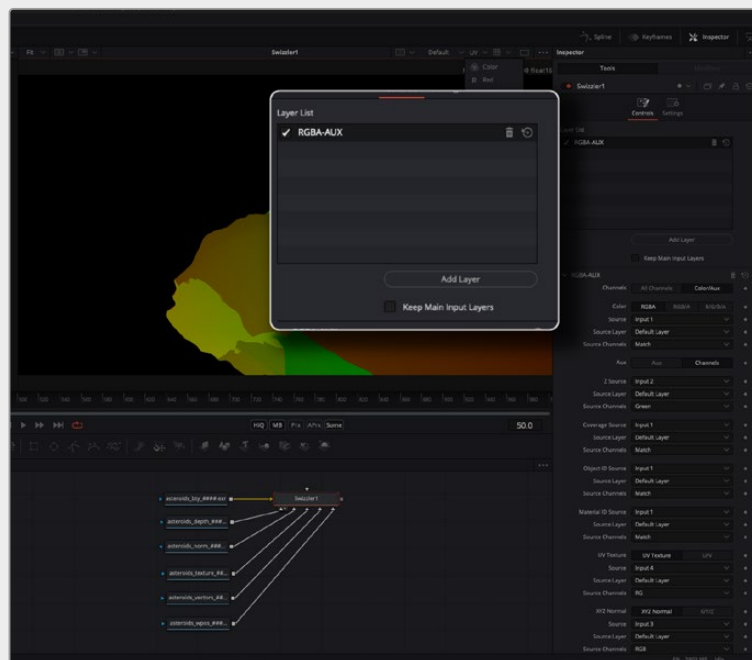
Assigning Aux channels

In the following example we have multiple RGB sources. Each one is a pre-rendered pass which we want to assign to an Aux channel.



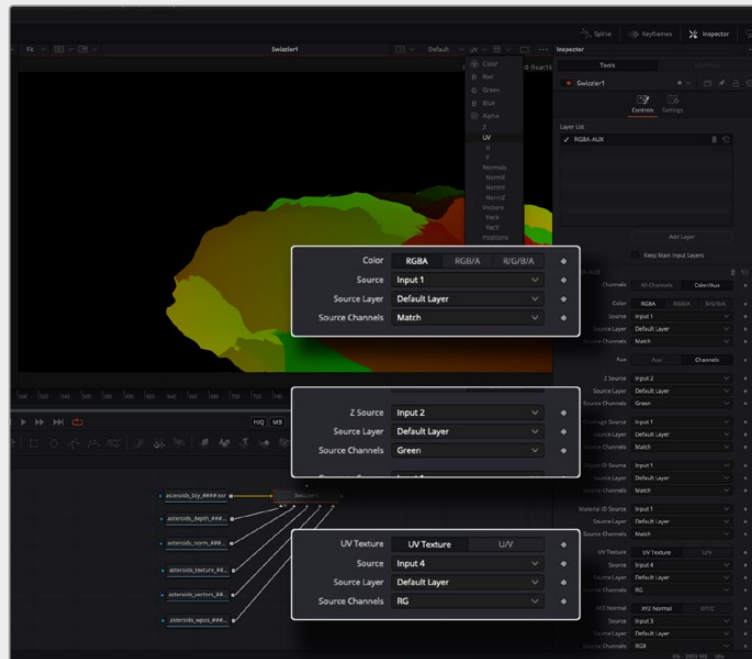
Connecting multiple RGB sources to the Swizzler

Each RGB pass is connected to a new input on the Swizzler.



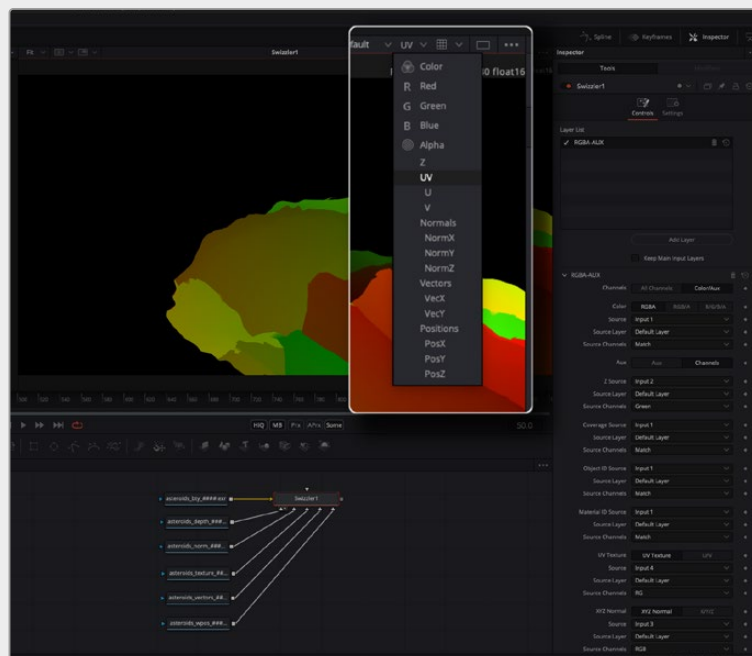
Creating a new layer in the Swizzler

A new layer has been added to the layer list, as there is only one layer, this will become the new 'default layer'



Assigning sources in the channel controls for the new layer

In the channel controls for the new layer, assign the appropriate source for each of the Aux channels.

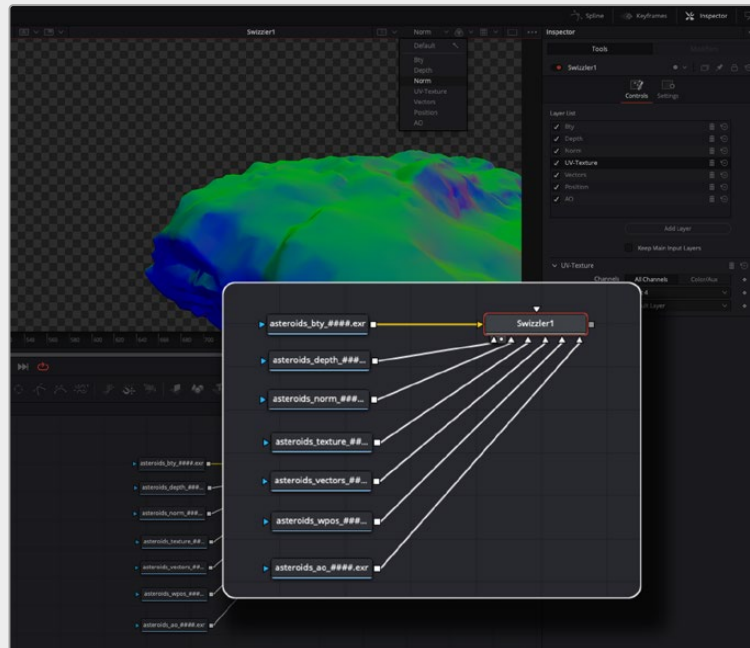


Previewing the new channels in the viewer

Once each of the Aux channels has been assigned an input, the channels can be previewed in the viewer.

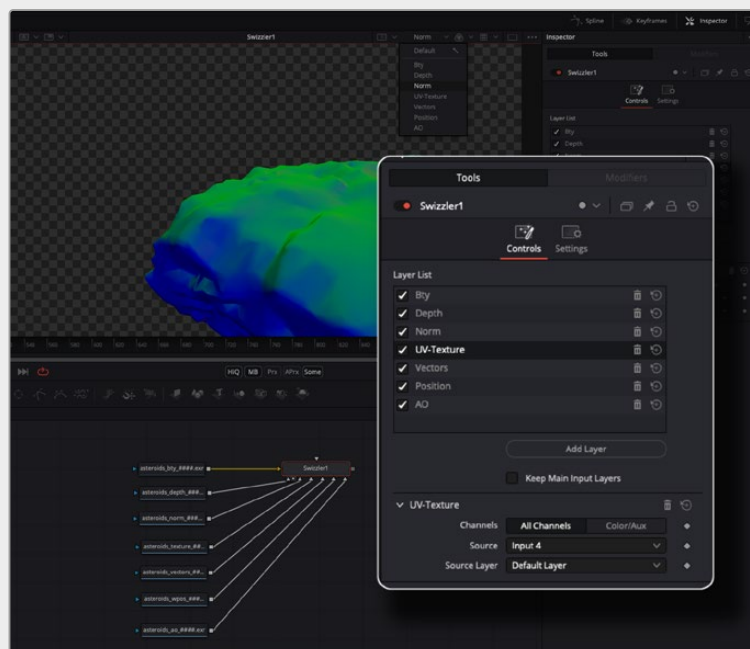
Creating a multilayer clip

In this example, we have multiple RGB sources. Each one is a pre-rendered pass that we want to assign to a new layer, creating a multilayer clip.



Connecting multiple RGB sources to the Swizzler

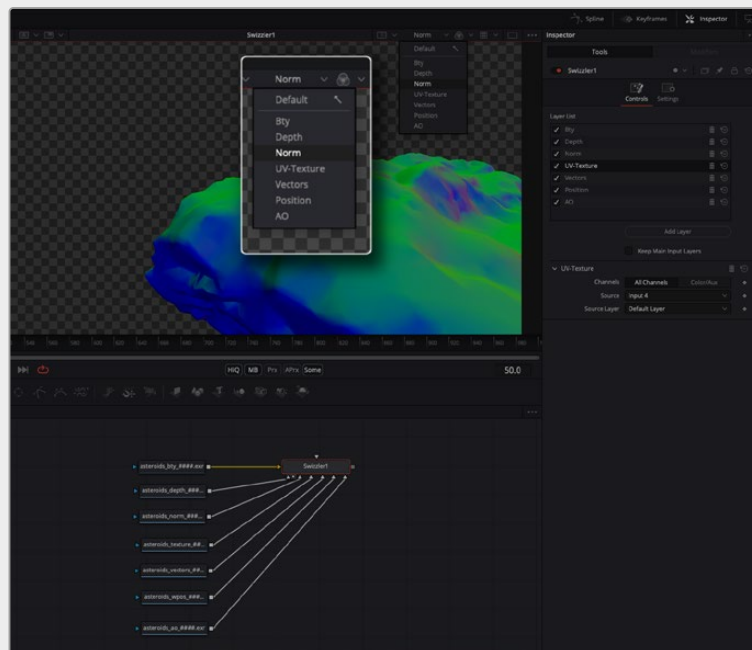
Each RGB pass is connected to a new input on the Swizzler.



Creating new layers in the Swizzler

In the layer list, add a new layer for every pass and give it a custom name. Select a layer in the layer list, and in the controls assign the appropriate source input for that layer.

For example, the UV pass has been connected to Input 4; selecting the custom layer labeled “UV-Texture” and choosing Input 4 from the Input list will push all the channels from Input 4 to the “UV-Texture” layer.



Selecting the new layers in the viewer

Once each of the inputs has been assigned a custom layer, these can be previewed in the viewer and used in the comp.

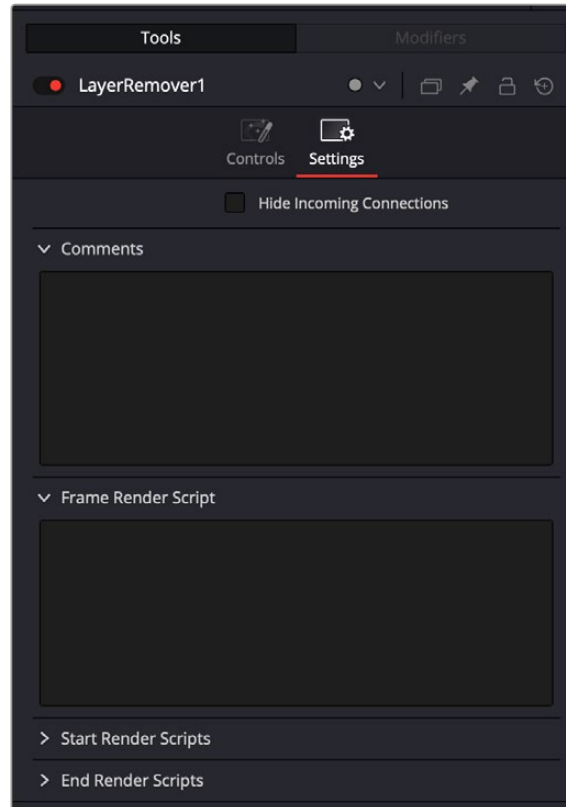
Common Controls

Settings Tab

The Settings tab in the Inspector is also duplicated in other Layer nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

The Common Controls

Settings Tab



The Settings tab

The Common Settings tab can be found on most tools in Fusion. The following controls are specific settings for Deep Image nodes.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node are displayed in the Inspector. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comments Tab

The Comments tab contains a single text control that is used to add comments and notes to the tool. When a note is added to a tool, a small red dot icon appears next to the setting's tab icon and a text bubble appears on the node. To see the note in the Node Editor, hold the mouse pointer over the node for a moment. The contents of the Comments tab can be animated over time, if required.

Scripting Tabs

The Scripting tab is present on every tool in Fusion. It contains several edit boxes used to add scripts that process when the tool is rendering. For more details on the contents of this tab, please consult the scripting documentation.